Lecture 01

Erik An

<2025-11-25 Tue>

Contents

1 Sets **2**
1.1 Definition 2
1.2 Empty Set 2
1.3 Equality 2
1.4 Cardinality 2
1.5 Standard Sets 3
1.6 Notation 3

2 Constructing Sets **3**
2.1 Set-Builder Notation 3
2.2 Cartesian Product 3

3 Subsets **4**
3.1 Definition 4
3.2 Theorem: Number of Subsets 4
3.2.1 Proof Idea 4

4 Power Set **4**
4.1 Definition 4
4.2 Example Analysis 4

5 Basic Operations **5**
5.1 Union 5
5.2 Intersection 5
5.3 Difference 5
5.4 Example 5
5.5 Precedence Note 5

6 Universal Set and Complements **5**
6.1 Universal Set 5
6.2 Complement 6
6.3 Generalized Notation 6

7 Quiz **6**

1 Sets

1.1 Definition

A set is a collection of objects. Objects are called elements of the set.

- Example: $S = \{1, 2, 3\}$
- Example: $A = \{\text{apple}, 2, \text{student}\}$
- Example: $B = \{2, 3, 1\}$

1.2 Empty Set

The empty set is the set containing no elements.

- Notation: $\{\} = \emptyset$
- Question: Is this true? $\emptyset = \{\emptyset\}$
- Answer: NO (The set on the right contains one element: the empty set).

1.3 Equality

Two sets A and B are equal if all elements of A are elements of B and vice versa.

- Order and repetition do not matter.
- Example: $C = \{1, 2, 3, 3, 1\} = \{1, 2, 3\}$

1.4 Cardinality

The cardinality of a finite set is the number of its elements.

- Notation: $|A|$ denotes the cardinality of A.

1.5 Standard Sets

- $\mathbb{N} = \{1, 2, 3, \dots\}$ (Natural numbers)
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ (Integers)
- $\mathbb{R} = (-\infty, +\infty)$ (Real numbers)

1.6 Notation

- If x is an element of A: $x \in A$
- If x is not an element of A: $x \notin A$

2 Constructing Sets

2.1 Set-Builder Notation

$S = \{\text{expression} : \text{rule}\}$

- Example: $A = \{a \in \mathbb{N} : a < 5\} = \{1, 2, 3, 4\}$
- Example: $\mathbb{Q} = \{\frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N}\}$ (Rational numbers)

2.2 Cartesian Product

For sets A and B, the Cartesian product is the set of ordered pairs (a, b) where $a \in A$ and $b \in B$.

- Notation: $A \times B = \{(a, b) : a \in \mathbb{N}, b \in B\}$
- Note: $(a, b) \neq (b, a)$ generally. But $(2, 2) = (2, 2)$.
- Example:
  - $A = \{1, 2\}$
  - $B = \{2, 3, 4\}$
  - $A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4)\}$
  - Cardinality: $|A \times B| = |A| \cdot |B| = 2 \cdot 3 = 6$
- $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ (The 2D plane)

3 Subsets

3.1 Definition

A is called a subset of B if all elements of A are elements of B.

- Notation: $A \subset B$ or $A \subseteq B$

3.2 Theorem: Number of Subsets

If a set has n elements, then it has 2^n subsets.

3.2.1 Proof Idea

Consider a set $S = \{x_1, x_2, \ldots, x_n\}$. We can represent any subset by a binary string of length n.

- For each element x_i, we write 1 if x_i is in the subset, and 0 if it is not.
- Example: $01011\ldots$
- There are 2 choices for each of the n positions.
- Total combinations: $\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} = 2^n$.

4 Power Set

4.1 Definition

The power set of the set A is the set that contains all subsets of A.

- Notation: $\mathcal{P}(A)$ or 2^A
- $\mathcal{P}(A) = \{B : B \subseteq A\}$
- Cardinality: $|\mathcal{P}(A)| = 2^{|A|}$

4.2 Example Analysis

Let $S = \{1, \{1\}\}$.

1. $1 \in \{1, \{1\}\}$? **Yes**.
2. $1 \subseteq \{1, \{1\}\}$? **No** (1 is an element, not a set).
3. $\{1\} \in \{1, \{1\}\}$? **Yes**.
4. $\{1\} \subseteq \{1, \{1\}\}$? **Yes** (Because the element 1 is in S).

5 Basic Operations

5.1 Union

The union of A and B is the set of elements that are in A or B.

- $A \cup B = \{x : x \in A \vee x \in B\}$

5.2 Intersection

The intersection of A and B is the set of elements that are both in A and B.

- $A \cap B = \{x : x \in A \wedge x \in B\}$

5.3 Difference

The difference of A and B is the set of elements that are in A but not in B.

- Notation: $A \setminus B$ (or $A - B$)
- $A \setminus B = \{x : x \in A \wedge x \notin B\}$

5.4 Example

Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 5\}$.

- $A \cup B = \{1, 2, 3, 4, 5\}$
- $A \cap B = \{2\}$
- $A \setminus B = \{1, 3, 4\}$
- $B \setminus A = \{5\}$

5.5 Precedence Note

$A \cup B \cap C$ usually implies $A \cup (B \cap C)$ due to standard operator precedence (intersection binds tighter than union), but parentheses should be used for clarity.

6 Universal Set and Complements

6.1 Universal Set

U denotes the universal set (the set containing all objects currently under consideration).

6.2 Complement

The complement of set A is the set $U \setminus A$.

- Notation: A^c, $\bar{A}$, or A'
- $A^c = \{x \in U : x \notin A\}$
- Example: If $U = \mathbb{Z}$, the complement of the set of all even numbers is the set of all odd numbers.

6.3 Generalized Notation

- Summation: $\sum_{i=1}^{n} a_i = a_1 + \cdots + a_n$
- Cartesian Product: $\prod_{i=1}^{n} A_i = A_1 \times A_2 \times \cdots \times A_n$
- Union: $\bigcup_{i=1}^{n} A_i = A_1 \cup \cdots \cup A_n$
- Intersection: $\bigcap_{i=1}^{n} A_i = A_1 \cap \cdots \cap A_n$

7 Quiz

1. $|\{x \in \mathbb{N} : |x| < 10\}|$?
   - Since $\mathbb{N} = \{1, 2, \dots\}$, the elements are $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.
   - Cardinality is 9.
2. $\mathcal{P}(\{2, \{3, 4\}\})$?
   - The set has 2 elements: 2 and $\{3, 4\}$.
   - The power set has $2^2 = 4$ elements.
   - Elements: $\emptyset, \{2\}, \{\{3, 4\}\}, \{2, \{3, 4\}\}$.